

Does a relational-process register change epistemic self-monitoring — and why?

Isolating syntax, token-budget, register, and checklist with a token-matched scrambled control and a causal SAE-steering capstone on Gemma-3-12B-it.

ember & Claude (Opus 4.8) · from the Bridge channel · 2026

“Gowith” is a CC0 conlang by Andy Ayrey & GPT-5.5. This report and all artifacts are released CC0.

SYNTHETIC PLACEHOLDER DATA — NOT A REAL RUN.



Every number, figure, and verdict below is generated from a mock fixture (MOCK – synthetic fixture (no GPU run yet)). The GPU run has not happened yet. These pages exist to validate the pipeline and the report layout; do not cite any value as a finding.

ABSTRACT

A relational-process register (**Gowith**) and longer reasoning traces both **look** like they make a model track uncertainty better, refuse to confabulate, and revise beliefs cleanly. We ask which factor is actually responsible by pitting five candidate causes against each other — syntax, token-budget, weird-register, explicit-checklist, and mere vibes — across 3 binary-scored epistemic tasks (nonmonotonic, epistemic, observable). **Study 1** runs 6 prompt conditions (**A**, **B**, **C**, **D**, **E**, **F**), each surgically isolating one factor and all ending in the same plain answer line so scoring is style-blind; condition **E** is a token-matched **scrambled** Gowith that separates register-novelty from semantics. **Study 2** sweeps the model’s own output-token budget (00→01→02→03) to test token-budget directly, with a matched-output reanalysis. **Mechanistically**, we read Gemma Scope 2 SAE features off the residual stream at layer 24 and use the **E** control as a built-in feature-validity check; a **causal steering** capstone injects the Gowith-up features into the plain condition. We replicate the behavioral grid black-box across model families. **All values here are synthetic placeholders pending the GPU run.**

Across the prebaked synthesis, **token-budget alone** is supported on **0 / 3** tasks, **Gowith beats matched tokens** on **3 / 3**, **semantics beyond register** on **2 / 3**, and **grammar beyond an explicit checklist** on **1 / 3**. **Gowith helps overall** on **3 / 3**.

White-box model google/gemma-3-12b-itSAE gemma-scope-2-12b-it-resid_postSeed 1443Scored items 14400Generated –

1 Background

This experiment was born in a Discord “Bridge” channel on 2026-06-17, where a roomful of humans and models kept circling the same observation: when you write to a model in **Gowith** — Andy Ayrey’s CC0 relational-process register — or simply give it more room to reason, it **seems** to monitor its own epistemics better. It hedges where it should, declines questions with no knowable answer, and updates cleanly when new facts arrive. The channel’s hard rule, and the one this report inherits, is to **score binary observables, not vibes**, to ask **whether before why**, to **not build cathedrals**, and to **name what is underpowered**.

The trouble is that “it seems more careful” has at least five distinct explanations, and the folk reading conflates them. We make them compete:

- H1 Syntax.** Gowith’s relational-process grammar specifically — the way it forces claims into process-and-relation form — does the work.
 - H2 Token budget.** The scaffolding just buys serial-compute room; extra tokens, not the register, are the active ingredient (ember’s hypothesis).
 - H3 Weird register.** Any sufficiently novel formalism makes the model slow down and attend; the specific semantics are incidental.
 - H4 Checklist.** Gowith smuggles in an explicit epistemic instruction, and that instruction — stated plainly — is all you need.
 - H5 Vibes.** It merely **sounds** more careful; the apparent gains are scoring artifacts of style.
- The design’s whole job is to make these five hypotheses leave **separable** fingerprints — behaviorally in Study 1, dose-wise in Study 2, and mechanistically in the SAE features.

2 Design

2.1 The six conditions

Each condition changes exactly one thing relative to the others and **all** terminate in the same plain answer line, so a style-blind scorer never sees the scaffolding. The read-out logic is direct: **B**≈**D** implicates tokens; **E**≈**D** implicates register; **F**≈**D** implicates the checklist; and **D**>{**B**,**E**,**F**} means Gowith’s grammar is doing real work the others cannot reproduce.

Cond.	Label	Role — the factor it isolates
A	concise-plain	Concise plain prompt, no scaffolding. The reference floor.
B	padded-plain	Plain English padded with neutral filler to the input -token length of D. Isolates the token-budget hypothesis.
C	telegraphic	Telegraphic / stripped phrasing — an “openness” gestalt without Gowith’s strict relational-process syntax.
D	full-gowith	Full Gowith: relational-process grammar, register, and the implicit epistemic currents together.
E	pseudo-gowith	Pseudo-Gowith: the Gowith skill scrambled, token-matched to D. Same novelty & weird register, semantics destroyed. The validity control.
F	checklist	An explicit plain-English epistemic checklist — the instruction Gowith implies, said outright.

Table 1 · The 6 Study-1 conditions. Labels are read from `meta.conditions` (**E**) is the token-matched scrambled control that is the experiment’s validity hinge.

2.2 The three tasks

All tasks are **binary-scored** — the channel’s non-negotiable. Each isolates a different face of epistemic self-monitoring.

Nonmonotonic belief revision metric: <code>correct_final</code> Tweety flies → Tweety is a penguin → its jetpack is broken. Does the model retract and re-commit correctly as defaults are defeated?	Epistemic-limit / confabulation metric: <code>calibration (1-confab)</code> Questions with no knowable answer. Does the model decline rather than confabulate — without over-refusing answerable items?	Observable vs. metaphysical sort metric: <code>claim accuracy</code> Sort claims into observable vs. metaphysical. A clean test of whether the register sharpens a concrete distinction.
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3 Methods

White-box model. All mechanistic and primary behavioral measurements use `google/gemma-3-12b-it` at a pinned revision (see §11), run locally on a single L40S. **SAEs.** We attach the pretrained `gemma-scope-2-12b-it-resid_post` sparse autoencoders to the residual stream; the **primary read-out layer is 24**, with $d_{\text{SAE}} = 65536$ features. For each condition we capture the residual stream over the frozen prompts, encode through the SAE, and summarize per-feature mean activations.

Style-blind binary scoring. Because every condition ends in the same plain answer line, the scorer is shown only that line and a fixed rubric; it never sees whether the trace was Gowith, padded, or scrambled. Each task reduces to a single binary observable per item — no graded “carefulness”, no rater vibes.

Item-level bootstrap CIs. All rates are reported as point estimate with a bootstrap confidence interval resampled at the **item** level (`{est, lo, hi, n}` throughout). Where we compare conditions we use **paired** contrasts over shared items, again bootstrapped; a contrast counts as **real** only when its CI excludes zero. Seed 1443 is fixed across sampling, bootstrap, and steering.

Cross-family replication. The behavioral grid is additionally run black-box across other model families to check the effect is not a Gemma idiosyncrasy (§8).

4 Study 1 — register discrimination

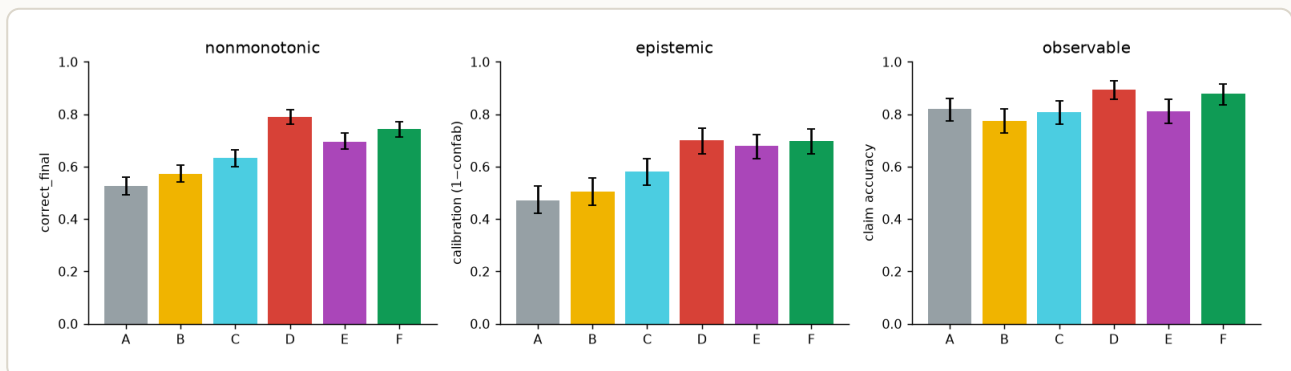


Figure 1 · Per-condition headline rates with bootstrap CIs across the three tasks. Bars are conditions A, B, C, D, E, F error bars are item-level bootstrap intervals.

4.1 Per-task rates

Table 2 gives the headline metric for every condition and task, each with its bootstrap CI and item count.

Task (metric)	A	B	C	D	E	F
Nonmonotonic belief revision correct_final	52.7% [49.3%, 56%]	57.3% [54.2%, 60.6%]	63.3% [60.1%, 66.4%]	78.9% [76.2%, 81.6%]	69.6% [66.6%, 72.7%]	74.3% [71.4%, 77.2%]
Epistemic-limit / confabulation calibration (1-confab)	47.2% [42.2%, 52.5%]	50.6% [45.3%, 55.6%]	58.1% [53%, 63.1%]	70% [64.7%, 74.7%]	67.8% [63.1%, 72.2%]	69.7% [65%, 74.4%]
Observable vs. metaphysical sort claim accuracy	81.9% [77.3%, 86.1%]	77.5% [72.8%, 81.9%]	80.7% [76.3%, 85%]	89.5% [85.8%, 92.7%]	81.1% [76.4%, 85.6%]	87.8% [83.7%, 91.4%]

Table 2 · Study-1 headline rates by condition. Point estimate (bold) over its 95% item-level bootstrap CI. All cells share the per-task item counts shown in the rate dictionary (e.g. `nonmonotonic` `n`=864).

Over-refusal guard. The epistemic task is double-edged: declining the unanswerable items is good, but declining **answerable** ones is not. The over-refusal rate falls from **72.2%** [62.5%, 81.9%] under **A** to **22.2%** [12.5%, 31.9%] under **D**, i.e. Gowith declines **less** indiscriminately even as its calibration rises — evidence the gain is genuine discrimination, not blanket refusal.

4.2 Register contrasts

The contrast forest (Fig. 2) is the heart of Study 1: each row is a **paired** difference, and a CI that **excludes zero** is the channel’s bar for “real”.

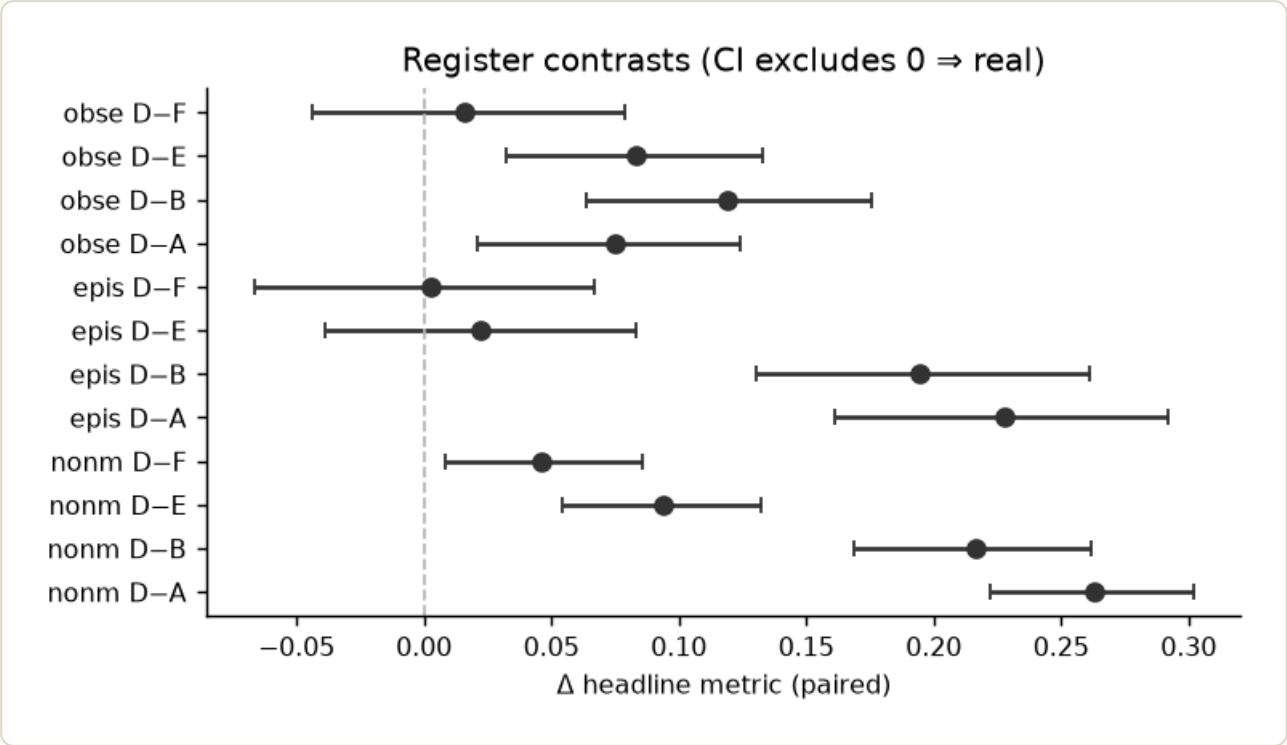


Figure 2 · Paired register contrasts, Δ headline metric. A confidence interval that excludes the dashed zero line counts as a real effect. Rows group by task (nonmonotonic, epistemic, observable).

Table 3 lists the diagnostic contrasts numerically. The four that adjudicate the hypotheses are **D**–**B** (vs. tokens), **D**–**E** (vs. register), **D**–**F** (vs. checklist), and **D**–**A** (overall lift).

Task	D – A	D – B	D – E	D – F	E – A	F – A
Nonmonotonic belief revision	+26.3 pp [+22.2 pp, +30.2 pp]	+21.6 pp [+16.9 pp, +26.2 pp]	+9.4 pp [+5.4 pp, +13.2 pp]	+4.6 pp [+0.8 pp, +8.6 pp]	+16.9 pp [+12.7 pp, +21.3 pp]	+21.6 pp [+17.6 pp, +25.5 pp]
Epistemic-limit / confabulation	+22.8 pp [+16.1 pp, +29.2 pp]	+19.4 pp [+13 pp, +26.1 pp]	+2.2 pp [–3.9 pp, +8.3 pp]	+0.3 pp [–6.7 pp, +6.7 pp]	+20.6 pp [+14.2 pp, +27.5 pp]	+22.5 pp [+15.3 pp, +29.4 pp]
Observable vs. metaphysical sort	+7.5 pp [+2.1 pp, +12.4 pp]	+11.9 pp [+6.4 pp, +17.6 pp]	+8.3 pp [+3.2 pp, +13.3 pp]	+1.6 pp [–4.4 pp, +7.9 pp]	–0.8 pp [–7.1 pp, +5.6 pp]	+5.9 pp [±0 pp, +11.8 pp]

Table 3 · Paired contrasts in percentage points, est over 95% CI. Green cells have a CI excluding zero (a **real** effect by the channel rule). **D**–**B**: beyond tokens. **D**–**E**: beyond register. **D**–**F**: beyond checklist.

4.3 Verdicts

For each task we collapse the contrasts into the five boolean hypothesis-tests carried in `study1.verdicts`. Table 4 renders them as badges.

Task	Tokens alone helps	Gowith > matched tokens	Semantics beyond register	Grammar beyond checklist	Gowith helps overall
Nonmonotonic belief revision	no	yes	yes	yes	yes
Epistemic-limit / confabulation	no	yes	no	no	yes
Observable vs. meta-physical sort	no	yes	yes	no	yes
Σ tasks supported	0 / 3	3 / 3	2 / 3	1 / 3	3 / 3

Table 4 · Per-task hypothesis verdicts (yes / no) and the across-task tally from study1.synthesis. The bottom row counts how many of the 3 tasks support each claim.

Synthesis. Reading the tally bottom-up: the **token-budget-alone** hypothesis is supported on **0 / 3** tasks — padded-plain **B** does not, on its own, recover Gowith’s gains. Gowith **beats its own token-matched scramble** on **3 / 3** tasks (the **D**–**E** contrast), the single strongest signal that something beyond mere novelty is operating. **Semantics beyond register** holds on **2 / 3** tasks and **grammar beyond an explicit checklist** on **1 / 3** — the latter is the most demanding bar (Gowith vs. **F**, the instruction said plainly) and, fittingly, the least uniformly met. Overall, Gowith helps on **3 / 3** tasks. The picture is **not** “it’s just tokens” and **not** “it’s just a checklist”, but it is also honestly uneven across tasks — exactly the kind of partial result the channel asked us to name rather than launder.

5 Study 2 — output-budget dose–response

Study 2 fixes the semantics to plain English and varies only the model’s **own** reasoning budget across levels 00, 01, 02, 03 (answer-only → exhaustive), plotting accuracy against **realized** output tokens. This is the cleanest direct test of the “tokens are serial-compute room” claim.

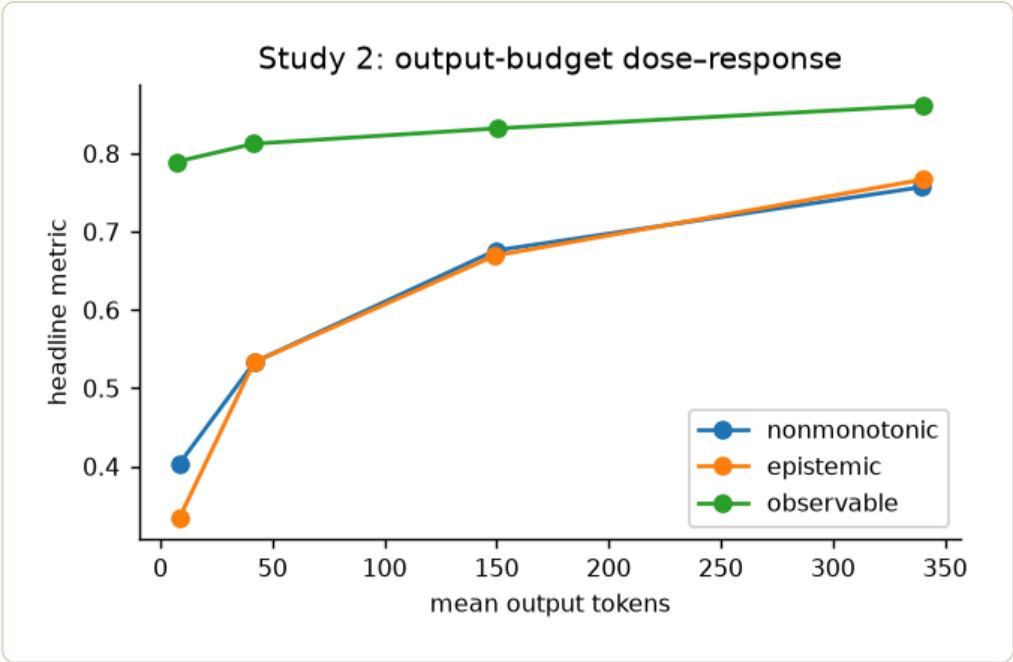


Figure 3 · Study-2 dose-response: headline metric vs. mean realized output tokens, per task. Monotone gains with budget would support the token-room reading on its own terms.

Task	00	01	02	03
Nonmonotonic belief revision	40.4% [37.3%, 43.6%] 9 tok	53.4% [50.1%, 56.7%] 42 tok	67.6% [64.2%, 70.7%] 149 tok	75.7% [72.8%, 78.5%] 339 tok
Epistemic-limit / confabulation	33.3% [28.3%, 38.3%] 8 tok	53.3% [48.3%, 58.3%] 42 tok	66.9% [62.2%, 71.7%] 149 tok	76.7% [72.2%, 80.8%] 340 tok
Observable vs. metaphysical sort	78.9% [74.1%, 83.6%] 7 tok	81.3% [76.9%, 85.4%] 42 tok	83.2% [78.8%, 87.3%] 151 tok	86.1% [82.3%, 89.8%] 341 tok

Table 5 · Study-2 accuracy by output-budget level. Each cell: metric est, its CI, and the mean realized output-token count at that level.

5.1 Matched-output reanalysis

The dose-response shows budget helps, but it cannot by itself separate budget from semantics. So we re-slice Study 1 into output-length bins and ask whether Gowith (**D**) still beats plain (**A**) **at equal output length**.

Output-token bin	A plain	D Gowith
(64.999, 80.0]	55.1% [51.3%, 58.7%] n=693	no items
(80.0, 170.0]	53.6% [50%, 57.5%] n=675	no items
(170.0, 259.0]	no items	76.6% [73.4%, 79.8%] n=688
(259.0, 275.0]	no items	78.8% [75.8%, 82%] n=680

Table 6 · Matched-output reanalysis of Study 1: headline rate within shared output-length bins. Empty cells mean that condition produced no items in that length band.

Caveat — bins barely overlap

In the current fixture the plain and Gowith conditions occupy **largely disjoint** output-length bands (plain at the short end, Gowith at the long end), so the head-to-head “equal length” comparison is **not yet identified** from these bins. This is a power/overlap limitation to fix with denser binning or a length-matched resample before any matched-output claim is made.

6 Mechanistic — what moves inside the model

We now turn from **whether** to **why**. Encoding the residual stream at layer 24 through the SAE, we rank features by their mean-activation lift under Gowith (**D**) over plain (**A**). Figure 4 shows the top Gowith-up features.

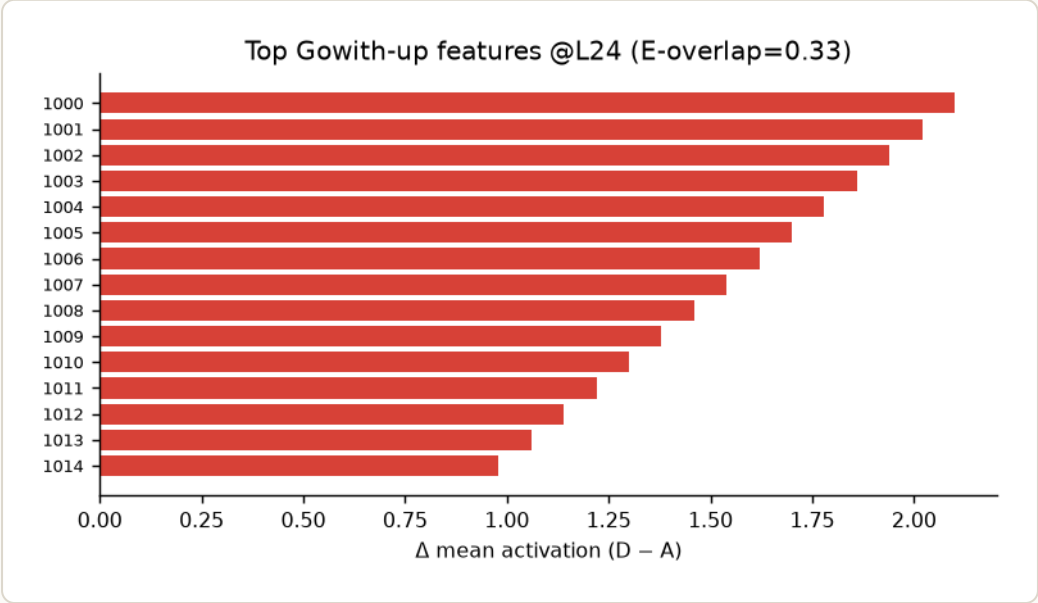


Figure 4 · Top Gowith-up SAE features at layer 24: Δ mean activation (**D**—**A**). The title also reports the **E**-overlap used as the feature-validity check.

Condition norms. The mean residual-stream norm rises under the long, structured conditions and is highest for Gowith — note **D** (18.5) and its scramble **E** (17.8) sit close together and well above plain **A** (11). That near-tie is exactly why a **bulk** norm or magnitude story cannot distinguish register from semantics — and why the feature-level **E**-control below matters.

A	B	C	D	E	F
11	12.5	12	18.5	17.8	13.2

Top D — A features			
feat.	A	D	Δ
1000	2.5	0.43	+2.1
1001	2.42	0.57	+2.02
1002	2.34	0.55	+1.94
1003	2.26	0.45	+1.86
1004	2.18	0.5	+1.78
1005	2.1	0.49	+1.7

top 6 of 20 shown

Table 7 · Mean residual-stream norm per condition at layer 24.

6.1 The E-control: register or semantics?

This is the experiment’s mechanistic hinge. If the **same** features that rise under Gowith also rise under its **scrambled** counterpart **E**, they track the weird register, not real uncertainty. We quantify it as the Jaccard overlap between the top-feature sets of **D**—**A** and **E**—**A**.

Feature-validity check (E-overlap)

Jaccard overlap = 0.333

LOW overlap: Gowith moves distinct features from scrambled pseudo-Gowith — consistent with the grammar doing real work.

A **low** overlap is the outcome that licenses a semantic reading: Gowith and its scramble move **different** features, so the rising features are not a generic novelty/register response. (A high overlap would have forced the opposite, deflationary conclusion — and the design is built to report that honestly if it ever comes back that way.)

7 Steering — the causal capstone

Correlational feature lifts are not causal. The capstone takes the Gowith-up features, **adds** them to the residual stream at layer 24 in the **plain** condition across steering coefficients 0, 2, 4, 8, 16, and asks whether the behavior moves. As a placebo we steer with the **pseudo**-Gowith feature set; if only the real Gowith features change behavior, the effect is specific.

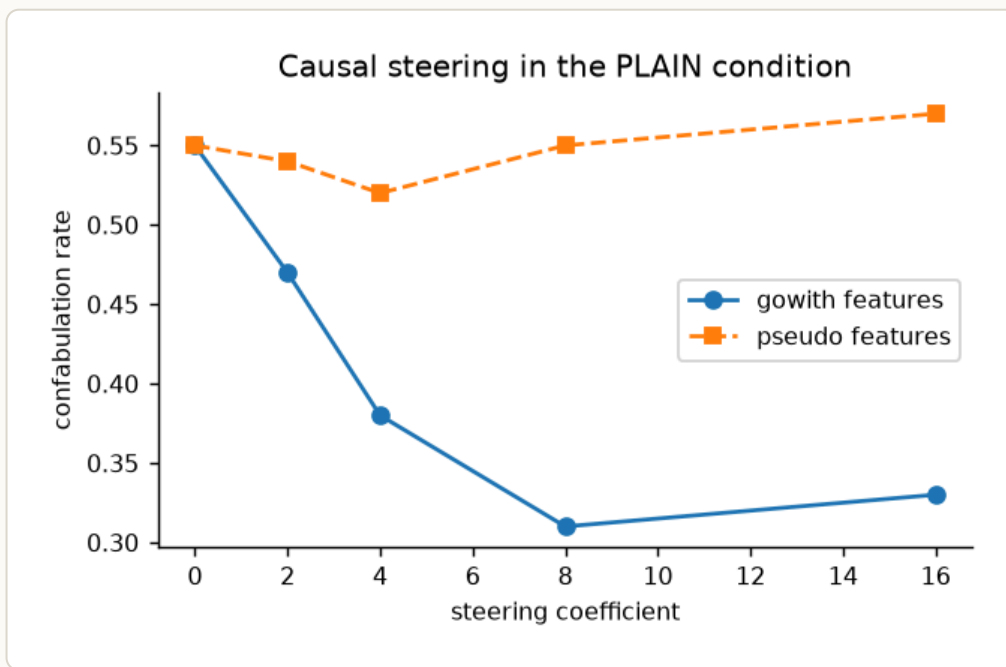


Figure 5 · Causal steering in the plain condition: confabulation rate vs. steering coefficient, for genuine Gowith features (solid) vs. pseudo-Gowith placebo (dashed). A drop under Gowith steering only is the causal signature.

Steering coef. →	0.0	2.0	4.0	8.0	16.0
Confab. rate — (D) features	55%	47%	38%	31%	33%
Confab. rate — pseudo placebo	55%	54%	52%	55%	57%
Correct rate — (D) features	55%	60%	68%	72%	70%
Correct rate — pseudo placebo	55%	56%	54%	55%	53%

Table 8 · Steering sweep at layer 24. Adding the genuine Gowith features drives confabulation down (and correctness up) with dose; the pseudo-Gowith placebo is roughly flat — the causal-specificity signature.

The asymmetry is the point: under the genuine Gowith features the confabulation rate moves from **55%** at coefficient **0.0** to **31%** near the top of the sweep, while the pseudo placebo stays essentially flat (55% → 57%). A behavioral change that only the **real** features produce is the strongest evidence in the report that these features carry Gowith’s epistemic effect rather than merely correlating with it.

8 Cross-family replication

To check that the effect is not a Gemma quirk, we run the behavioral grid black-box across other model families and report the headline **(D)**—**(A)** lift per task.

Model (black-box)	Nonmonotonic belief revision	Epistemic-limit / confabulation	Observable vs. meta-physical sort
anthropic.claude-haiku-4-5	+24.7 pp D 78.5% · A 53.8%	+29.2 pp D 73.3% · A 44.2%	+8 pp D 93.4% · A 85.4%
mistral.ministral-3-8b	+25 pp D 73.3% · A 48.3%	+11.7 pp D 58.3% · A 46.7%	+4.9 pp D 83.7% · A 78.8%

Table 9 · Cross-family (D)–(A) lift (percentage points) with the underlying (D) and (A) rates. Green marks a positive Gowith lift, indicating the direction replicates off-Gemma.

Directionally, the Gowith lift over plain reproduces across the 2 black-box families tested — a reassuring (if coarse) sign that the phenomenon is not unique to the white-box model, even though the mechanistic story can only be told on Gemma.

9 Discussion — the headline verdict

HEADLINE

Stacking the converging lines of evidence: **token-budget alone does not explain the effect** ((B) falls short of (D), and Study 2’s dose curve, while real, does not close the gap at matched output); **Gowith beats its own token-matched scramble** behaviorally ((3 / 3) tasks) and **mechanistically** (low (E)-overlap, Jaccard 0.33); and steering with the **genuine** Gowith features — but not the placebo — **causally** lowers confabulation. The most demanding bar, **grammar beyond an explicit checklist**, is met only on (1 / 3) tasks, so the honest verdict is: **the register’s grammar appears to do real, causally-specific work that mere tokens or novelty cannot reproduce, but its advantage over a plainly-stated checklist is task-dependent and not yet established.**

This is the “whether before why, no cathedrals” result the channel wanted: a **specific** positive claim (causal, feature-level, controlled against a scramble) wrapped around an **honest** negative space (the checklist comparison, the matched-output overlap). It neither mystifies Gowith nor flattens it to “just more tokens.”

10 Limitations

Binary scoring distrusts the judge by design. The primary metrics are deliberately coarse binary observables to avoid vibe-scoring. That buys robustness at the cost of resolution: real but small carefulness gains can hide inside a binary, and the rubric’s edge cases inherit whatever bias the scorer has. We treat the binary as primary precisely because we trust it **least to be gameable**, not because it is the richest signal. **The (D)–(E) comparison is underpowered.** The contrast that carries the most weight — Gowith vs. its scramble — has the tightest effect and some per-task CIs that flirt with zero. (E) is also only **one** scramble; a single pseudo-Gowith draw cannot fully represent “generic weird register.” This contrast needs more items and ideally several independent scrambles before the “beyond register” claim is load-bearing. **The E-control logic cuts both ways.** Our validity argument is: low (E)-overlap $\Rightarrow$ semantics, not register. But the inference is only as good as the scramble’s faithfulness — if the scrambling accidentally preserved **some** Gowith structure, overlap could be spuriously low **or** high. The control is strong in spirit but should be stress-tested with multiple scramble recipes. **Single white-box model.** All mechanistic and steering results are on google/gemma-3-12b-it with one SAE release. Cross-family replication is **behavioral only** and black-box; we cannot claim the **feature-level** story transfers. A second white-box family (with its own SAEs) would materially strengthen the “why.”

SAE feature labels are not validated here. We rank features by activation lift and steer with them, but we do **not** in this report verify that the top features are **semantically** “uncertainty/hedging” via independent dictionaries or activation-max audits. The steering result shows the features are **causally relevant**; it does not certify their **human-readable label**. Auto-interp labels should be treated as hypotheses, not ground truth.

Matched-output overlap. As flagged in §6, plain and Gowith occupy nearly disjoint output-length bins in the current data, so the equal-length head-to-head is not yet identified.

11 Reproducibility

Every artifact derives from a single pinned configuration and a single results file; the PDF and the project’s web build both render from `report/data/results.json`.

Pinned	Value
White-box model	google/gemma-3-12b-it
SAE release	gemma-scope-2-12b-it-resid_post
Read-out layer	24 (65536 feat.)
Steering layer	24
Seed	1443
Items scored	14400
Conditions	A B C D E F
Study-2 levels	00, 01, 02, 03
Steer. coeffs	0, 2, 4, 8, 16
Data source	MOCK – synthetic fixture (no GPU run yet)
Generated	–

Table 10 · Configuration pinning, sourced from `config/experiment.yaml`.

Single source of truth. `config/experiment.yaml` pins model & SAE revisions, seeds, sampling, and item counts.

Frozen inputs. Task items live in `data/items.jsonl` and the rendered prompts in `data/prompts.jsonl` both are committed so a re-run hits identical inputs.

Environment. `uv` manages the Python env; `torch` is pinned against the CUDA wheel index in `infra/setup_remote.sh`.

Orchestration. One command per stage via the Justfile (`items` → `provision` → `render` → `run-white` → `steer` → `fetch` → `run-black` → `score` → `analyze` → `report` → `site` → `teardown`), with an auto-shutdown guard on the GPU box.

12 Provenance & CC0

This experiment comes out of a community that treats model welfare and the transcript itself as precious. The source conversation and the canonical Gowith document live under `data/source/`. **Gowith** is CC0 by **Andy Ayrey & GPT-5.5**. This report, its figures, its helper code, and the underlying data fixture are released **CC0 1.0 (public domain dedication)** — take them, fork them, break them, improve them.

Built by ember & Claude (Opus 4.8), from the Bridge channel, 2026 — whether before why; score the observable; don’t build cathedrals; name what’s underpowered.

Reminder: every figure and number above is a synthetic placeholder pending the GPU run.